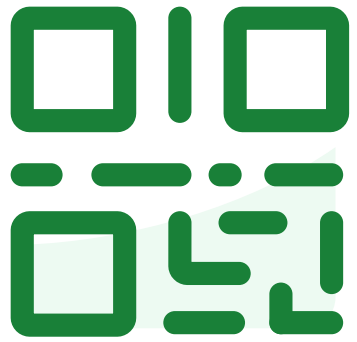


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Lecture 4

Probability and Density Estimation

Review of Probability and Fitting Probability Models to Data

EECS 189/289, Fall 2026 @ UC Berkeley

Joseph E. Gonzalez and Narges Norouzi

Alexa, call my
mom!



Calling your
mom!



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*Someone on YouTube just accidentally called their mom.

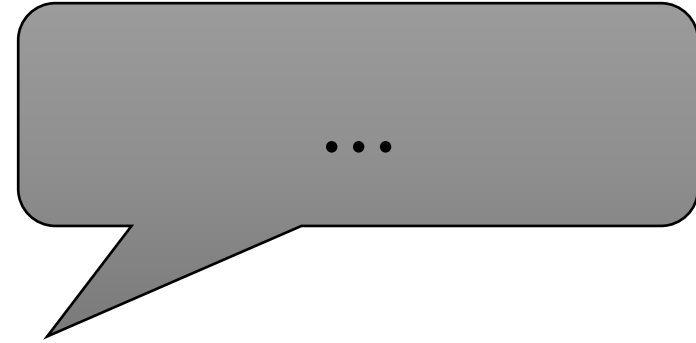


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Siri, call my
mom!



Siri



*Unlikely that anyone accidentally calls their mom...



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Try It Yourself

Take out your phone or laptop and say one of these out loud.

- “Hey Siri, what is the airspeed velocity of an unladen swallow?”
- “Alexa, set a timer for one second.”
- “OK Google, tell me a joke about Berkeley.”

Now say them again, but replace the first word: “**Hey Sardine...**”, “**Alaska...**”, “**OK Doodle...**”

Some devices woke up. Some did not. Some woke up for the wrong phrase.

- Every one of those devices was running a detector on **every second of sound in this room.**



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Siri

How does a smart assistant
“know when to listen”?

Wake Words

A **wake word** is a verbal cue that triggers voice assistants to start actively listening.

Example: *“Siri, set an alarm for 6:00AM”*

Most voice assistants **continuously** run a wake word detector model on every sound they hear.



Wake Words

A **wake word** is a verbal cue that triggers voice assistants to start actively listening.

Example: *“Siri, set an alarm for 6:00AM”*

Most voice assistants **continuously** run a wake word detector model on every sound they hear.



Building a Wake-Word Detector



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Is this a learning problem?

➤ **Yes** – difficult to describe easy to demonstrate

What kind of learning problem?



What kind of learning problem is wake word detection.

Building a Wake-Word Detector



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Is this a learning problem?

➤ Yes – difficult to describe easy to demonstrate

What kind of learning problem is this?

➤ Supervised (we will collect labeled examples)

➤ Binary classification (is this sound a wake word? Yes/No)

What kind of model should we use?

➤ Many possible solutions, ... probably a small neural network.

How accurate
does the
detector need to
be?

Building a Wake-Word Detector



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Streamed to the cloud
for processing.

For every segment of sound (e.g., 1 second):

- We observe a sound vector $\mathbf{X} \in \mathbb{R}^D$ (feature vector)
- Our model $f_w(\mathbf{X})$ (e.g., a Neural Network) predicts:

1: if a wake word was said

0: otherwise

Introduce **Random Variables** to
Model this Process

Building a Wake-Word Detector



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For every segment of sound (e.g., 1 second) there are **three random variables**:



We ultimately observe X and Y but today we will focus on the relationship between Z and Y (the *joint probability*).



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Today's Plan

Probability. Joint, marginal, conditional, independence, Bayes.

➤ The detector fires. What is the chance a wake word was said?

Expectations. Turning probabilities into quantities with units.

➤ What that accuracy costs per day.

Density estimation. Combining probability and optimization to fit models

➤ **Maximum likelihood:** the objective behind most models in this course.

➤ **Maximum a Posteriori:** accounting for priors.

A brief review of the

Basics of Probability



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The Joint Probability Distribution

Let X and Y be **discrete random variables** that take values:

$$X \in \{x_1, \dots, x_L\} \quad \text{and} \quad Y \in \{y_1, \dots, y_M\}$$

The joint probability function $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

The joint probability satisfies the **following two properties**:

$$\forall x_i, y_j: p(X = x_i, Y = y_j) \geq 0 \quad \text{(Non-negativity)}$$

$$\sum_{i=1}^L \sum_{j=1}^M p(X = x_i, Y = y_j) = 1 \quad \text{(Normalization)}$$

$p(X = x_i, Y = y_j)$	0	1	
0	0.2	0.1	0.3
1	0.25	0.45	0.7
	0.45	0.55	1



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The Joint Probability Distribution

Let X and Y be **discrete random variables** that take values:

$$X \in \{x_1, \dots, x_L\} \quad \text{and} \quad Y \in \{y_1, \dots, y_M\}$$

The joint probability function $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

The Sum Rule (Marginalization): defines the distribution over a subset of the random variables.

$p(X = x_i, Y = y_j)$	0	1	$p(X = x_i)$
0	0.2	0.1	0.3
1	0.25	0.45	0.7
$p(Y = y_j)$	0.45	0.55	1

$$p(X = x_i) = \sum_{j=1}^M p(X = x_i, Y = y_j)$$



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Conditional Probability

We define the conditional probability as the chance of observing $Y = y_j$ given **we observed** $X = x_i$

$$p(Y = y_j | X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)}$$

Example:

What is the $p(Y = y_1 | X = x_2)$?

$$\frac{\begin{array}{|c|c|} \hline x_2 & 0.25 \\ \hline \end{array} \quad 0.45}{0.7} = \begin{array}{|c|c|} \hline 0.36 & 0.64 \\ \hline \end{array} \quad p(Y = y_1 | X = x_3)$$

$p(X = x_i, Y = y_j)$	0	1	$p(X = x_i)$
0	0.2	0.1	0.3
1	0.25	0.45	0.7
$p(Y = y_j)$	0.45	0.55	1



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Product Rule: Chain Rule of Probability

Recall the definition of the conditional:

$$p(Y = y_j | X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)}$$

Multiply both sides by $p(X = x_i)$ you obtain the **product rule**:

$$p(X = x_i, Y = y_j) = p(Y = y_j | X = x_i) p(X = x_i)$$

The product rule is like a **chain rule** of probability, it allows us to **factorize** joint probability distributions into conditionals:

$$p(X, Y, Z) = p(Z | X, Y) p(X|Y) p(Y)$$

- *Notation*: we often drop x_i , y_j , and z_k for convenience.



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Bayes' Theorem

Named after Thomas Bayes who was an 18th century statistician, philosopher, and minister.

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \frac{p(B|A)p(A)}{\sum_{A'} p(B|A')p(A')}$$

- Derived from product rule: $p(A, B) = p(A|B)p(B) = p(B|A)p(A)$



Useful in Bayesian statistics:

- $p(A)$ is the **prior belief** about A
- $p(B|A)$ is the **likelihood** of an observation B given those beliefs
- $p(A|B)$ is our **posterior belief** about A given our observation B



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Independent Random Variables

Two variables X and Y are independent if the joint probability factorizes:

$$X \perp Y \Rightarrow p(X, Y) = p(X) p(Y)$$

Using the definition of conditionals:

$$X \perp Y \Rightarrow p(Y|X) = \frac{p(Y, X)}{p(X)} = \frac{p(Y)p(X)}{p(X)} = p(Y)$$

Independent and Identically Distributed (IID) data (X_i, Y_i) is:

$$p(X_1, Y_1, \dots, X_N, Y_N) = \prod_{n=1}^N p(X_n, Y_n)$$

- We will often assume IID data (though it is often not true in practice).



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Summarizing the Four Rules

Sum rule (marginalization): collapse a variable by summing it out.

$$p(X = x_i) = \sum_{j=1}^M p(X = x_i, Y = y_j)$$

Product rule (chain rule): factor a joint into a chain of conditionals.

$$p(X = x_i, Y = y_j) = p(Y = y_j | X = x_i) p(X = x_i)$$

Bayes' theorem: reverse the direction of a conditional.

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \frac{p(B|A)p(A)}{\sum_{A'} p(B|A')p(A')}$$

Independence: the special case where conditioning changes nothing.

$$X \perp Y \implies p(X, Y) = p(X) p(Y)$$

What happens with rare events?

Analyzing the Wake Word Detector

Building a Wake-Word Detector



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For every segment of sound there are **three random variables**:



Today we will focus on the relationship between Z and Y :

$$p(Z, Y) = p(Z) p(Y | Z)$$

Prior: $p(Z)$ the probability that a wake word is said

Likelihood: $p(Y | Z)$ the likelihood of detecting a wake word given whether a wake word was said.

Today we will focus on the relationship between Z and Y :

$$p(Z, Y) = p(Z) p(Y | Z)$$



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Prior: $p(Z)$ the probability that a wake word is said

Likelihood: $p(Y | Z)$ the likelihood of detecting a wake word given whether a wake word was said.

Our **wake word detector** can be characterized by:

➤ **Detection rate (True Positive Rate / Recall):** $p(Y = 1 | Z = 1)$

➤ If a wake word is said, do we detect it?

➤ **False alarm rate (False Positive Rate):** $p(Y = 1 | Z = 0)$

➤ If a wake word is not said, do we say that it was?



How could we improve the Wake Word Detector?

Building a Wake-Word Detector



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Today we will focus on the relationship between Z and Y :

$$p(Z, Y) = p(Z) p(Y | Z)$$

➤ **Detection rate (True Positive Rate / Recall):** $p(Y = 1 | Z = 1)$

➤ **False alarm rate (False Positive Rate):** $p(Y = 1 | Z = 0)$

We really want to know the **Precision:** $p(Z = 1 | Y = 1)$

➤ Depends on the **detector** and **rarity of wake words** $p(Z = 1)$

Ultimately want high **precision** and **recall**.



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Analyzing the Wake Word Detector

Let's assume wake words are rare and occur in about 0.01% of segments:

$$p(Z = 1) = 0.0001 \quad \text{and} \quad p(Z = 0) = 1 - p(Z = 1) = 0.9999$$

Suppose the wake word model **detects 99% of the wake words**:

$$p(Y = 1 | Z = 1) = 0.99$$

And **falsely detects (false positive)** a wake word just 0.1% of the time:

$$p(Y = 1 | Z = 0) = 0.001$$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(Z = 1 | Y = 1) = \frac{p(Y = 1, Z = 1)}{p(Y = 1)} = \frac{p(Y = 1 | Z = 1) p(Z = 1)}{p(Y = 1)}$$

Bayes' Theorem



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Analyzing the Wake Word Detector

We assumed rare wake words: $p(Z = 1) = 0.0001$

A **true positive rate (recall)** of 99%: $p(Y = 1 | Z = 1) = 0.99$

And a **false positive rate** of 0.1%: $p(Y = 1 | Z = 0) = 0.001$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(Z = 1 | Y = 1) = \frac{p(Y = 1 | Z = 1) p(Z = 1)}{p(Y = 1 | Z = 1) p(Z = 1) + p(Y = 1 | Z = 0) p(Z = 0)}$$
$$p(Z = 1 | Y = 1) = \frac{0.99 * 0.0001}{0.99 * 0.0001 + 0.001 * 0.9999} = 0.09 = 9\%$$

Is that good (enough)?

How would we improve it?



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Bayesian Updates: Wake Word Detector

Prior: we assumed rare wake words $p(Z = 1) = 0.0001$

After observing the test $Y = 1$ we **updated our belief** about the presence of the wake word using Bayes Rule:

$$p(Z = 1 | Y = 1) = \frac{p(Y = 1, Z = 1)}{p(Y = 1)} = \frac{p(Y = 1 | Z = 1) p(Z = 1)}{p(Y = 1)} = 0.09$$

Posterior: the updated belief $p(Z = 1 | Y = 1)$ after the observation.

Bayesian Updating: the process of starting from a **prior belief** and then using **observations** to compute **posterior distribution** (updated belief) is central to **Bayesian machine learning**.



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Revisiting the Math with Counts

Imagine 1,000,000 one-second segments.

- Wake words are said in 0.01% of them: **100 segments** contain a wake word, 999,900 do not.
- The detector catches 99% of the real ones: **99 true detections.**
- The detector fires on 0.1% of the silent ones: **999 false alarms.**

The detector fires $99+999 = 1,098$ times. **Only 99 of those are real.**

- $99 / 1,098 \approx 9\%$

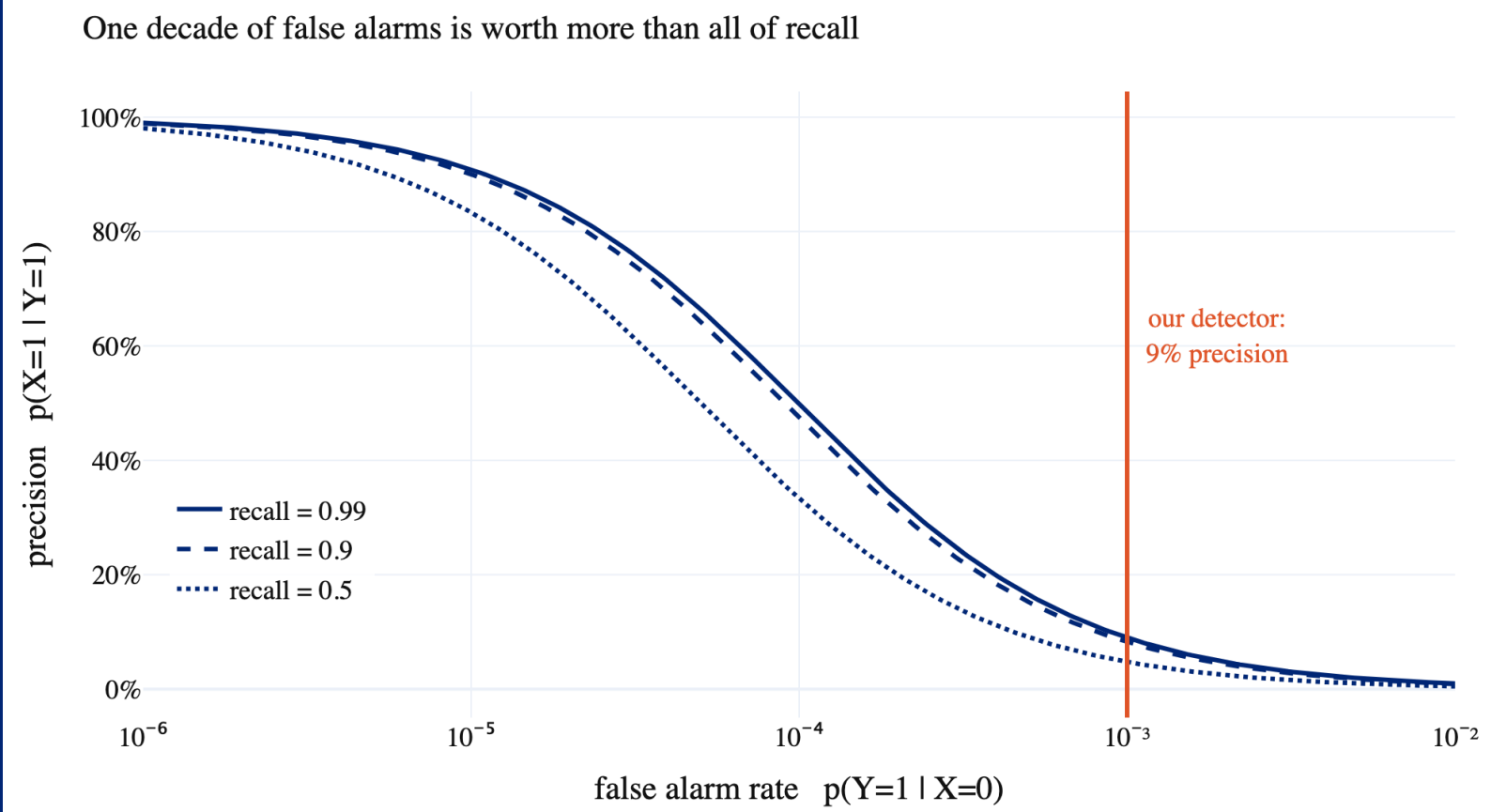
The base rate dominates. There are 10,000 times more silent segments than wake-word segments, so a **small error rate on a huge population** becomes a real challenge.



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Demo

Which knob matters?





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What Is 9% Precision Costing Us?

Every time the detector fires, the device **invokes the full speech model**. Invoking the full speech model can be expensive. Why?

- **Energy:** to **evaluate** the full model or **send the data** to the cloud
- **Money:** to evaluate the **model in a datacenter**.

Let C be the cost (e.g., \$ or Joules) of invoking the full speech model, what is the expected cost per day for false alarms?

- Keep in mind each segment of sound **is not independent**

To answer this, we will compute an **expectation**.



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Expectations

The **expectation of a random variable** X is the weighted average with respect to a probability distribution $p(x)$:

$$\mathbb{E}[X] = \sum_x x p(x)$$

An expectation maps a random variable to a number, but a conditional expectation is a function of the variable being conditioned on.

The **conditional expectation** of a random variable is a **weighted average** with respect to the **conditional distribution**:

$$\mathbb{E}[X \mid y] = \sum_x x p(x|y)$$



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Functions of Random Variables

The function of a random variable is also a random variable:

$$Y = f(X)$$

The **expectation of a function** $f(X)$ is the weighted average with respect to a probability distribution $p(x)$:

$$\mathbb{E}[f] = \mathbb{E}[f(X)] = \sum_x f(x) p(x)$$

The expectation of a function of multiple variables is defined over the joint distribution:

$$\mathbb{E}[f(X, Y)] = \sum_{x, y} f(x, y) p(x, y)$$



For any two correlated random variables X and Y



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Linearity of Expectation

For *any* two random variables X and Y :

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

- Does not require any assumptions about $p(X, Y)$

$$\begin{aligned}\mathbb{E}[aX + bY + c] &= \sum_x \sum_y (a x + b y + c) p(x, y) \\&= \sum_x \sum_y (a x p(x, y) + b y p(x, y) + c p(x, y)) \\&= a \sum_x x \sum_y p(x, y) + b \sum_y y \sum_x p(x, y) + c \sum_x \sum_y p(x, y) \\&= a \mathbb{E}[X] + b \mathbb{E}[Y] + c\end{aligned}$$



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Variance

The **variance** of a random variable defines its **spread**:

$$\text{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Using linearity of expectation, we can rewrite the variance as:

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

What is $\text{var}[aX + b]$?

$$\begin{aligned}\text{var}[aX + b] &= \underbrace{\mathbb{E}[(aX + b)^2]}_{\mathbb{E}[a^2X^2 + 2abX + b^2]} - \underbrace{\mathbb{E}[aX + b]^2}_{(a\mathbb{E}[X] + b)^2} \\ &= a^2\mathbb{E}[X^2] + \cancel{2ab\mathbb{E}[X]} + \cancel{b^2} \quad \quad \quad = a^2\mathbb{E}[X]^2 + \cancel{2ab\mathbb{E}[X]} + \cancel{b^2} \\ \text{var}[aX + b] &= a^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) = a^2 \text{var}[X]\end{aligned}$$



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Covariance

The covariance of two random variables defines the degree to which they vary together:

$$\text{cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

The covariance of two **random vectors** $\mathbf{x}$ and $\mathbf{y}$ is the matrix:

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T])] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

We will return to covariance in greater detail in future lectures.



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What Is 9% Precision Costing Us?

Let C be the cost (e.g., \$ or Joules) of invoking the full speech model, what is the expected cost per day for false alarms?

➤ Note: each segment of sound is **not independent**

If each segment of sound is **1 second long**, then there about $24*60*60=86,400$ segments in a day.

➤ Let I_t be the indicator variable that the segment t is a **false alarm**

$$\mathbb{E}[\text{Total Cost}] = \mathbb{E}\left[\sum_{t=1}^{86,400} C I_t\right] = C \sum_{t=1}^{86,400} \mathbb{E}[I_t] = C \sum_{t=1}^{86,400} p(Z = 0, Y = 1)$$
$$p(Z = 0, Y = 1) = p(Y = 1 \mid Z = 0) p(Z = 0) = 0.001 * 0.9999$$

$$\mathbb{E}[\text{Total Cost}] = 86.4 * C$$

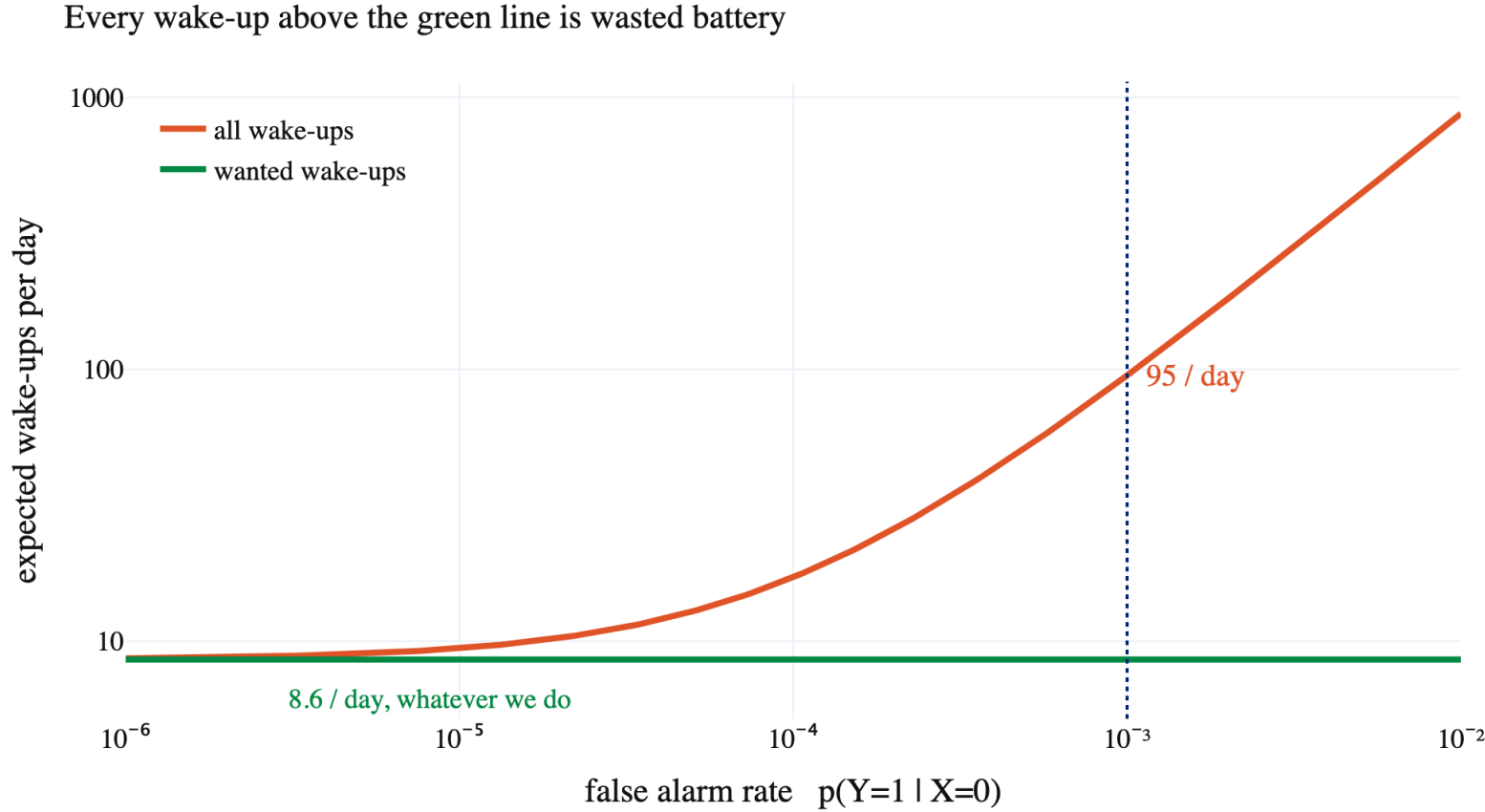
In lecture, I had incorrectly said $\mathbb{E}[I_t] = p(Z = 0 \mid Y = 1)$. This is **incorrect** since we want an indicator for all events not just when the detector triggered.



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Demo

What 9% precision
costs per day



From counting to

Modeling Distributions



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Modeling Distributions

Many machine learning models attempt to model the (joint) **probability distribution** of the **data**.

- Instead of modeling “Does this sound contain a wake word” they model “The probability that this sound is a wake word”.

There are **many ways** to model a (joint) probability distribution.

- Tabular representations
- Classic probability models
- Nonparametric models (histograms, kernel density estimation)



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Tabular Representations

Already saw a **tabular representation** of a **joint distribution**:

	y_1	y_2
x_1	0.2	0.1
x_2	0.1	0.1
x_3	0.15	0.35



$$p_w(X = x_i, Y = y_j) =$$

	y_1	y_2
x_1	$w_{1,1}$	$w_{1,2}$
x_2	$w_{2,1}$	$w_{2,2}$
x_3	$w_{3,1}$	$w_{3,2}$

Here p_w is a **parametric probability model** and $w \in \mathbb{R}^{3,2}$ satisfies:

$$\forall x_i, y_j: w_{i,j} \geq 0 \text{ (Non-negativity)} \quad \sum_{i=1}^L \sum_{j=1}^M w_{i,j} = 1 \text{ (Normalization)}$$

For **discrete distributions**, the tabular representation is **complete**: can express any discrete joint distribution

➤ Often not efficient (can be exponential in variables)

Bernoulli Distribution



The distribution of binary random variable $X \in \{0,1\}$ defined by:

$$X \sim \text{Bern}(\mu) \implies p(x|\mu) = \mu^x(1 - \mu)^{1-x}$$

where the parameter $0 \leq \mu \leq 1$.

$$\mathbb{E}[X] = 0(1 - \mu) + 1\mu = \mu$$

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mu - \mu^2 = \mu(1 - \mu)$$

$$\mathbb{E}[X^2] = 0^2(1 - \mu) + 1^2\mu = \mu$$

Note that for $\mu \notin \{0,1\}$, X never takes the expected value!

The Bernoulli distribution is used extensively in classification.



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Continuous Random Variables

For a **continuous random variable** X we use a **probability density** $p(x)$. We can use the density to compute the probability that X is between a and b :

$$P(a \leq X \leq b) = \int_a^b p(x) dx$$

A density $p(x)$ has the standard properties:

- **Normalization** $\int p(x) dx = 1$
- **Non-negativity** $p(x) \geq 0$.

A density is **not** a probability:

- $p(x)$ can be larger than 1,
- $P(X = x) = 0$ for any x .

Previous rules transfer with sums replaced by integrals:

- **Sum Rule:** $p(x) = \int p(x, y) dy$
- **Product Rule:** $p(x, y) = p(y | x) p(x)$
- **Expectation of a Function:** $\mathbb{E}[f(x)] = \int f(x) p(x) dx$

Fitting distributions to data with

Density Estimation and MLE



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Density Estimation

Density estimation is the process of trying to **infer the probability distribution $p(X)$** given observations (data) $\mathcal{D} = \{x_1, \dots, x_N\}$.

There are *infinitely many* density functions from which our data could have been sampled.

- Could be any function that assigns **non-zero probability to the data**
- The choice of function is a **modeling decision (inductive bias!)**

Density estimation is a fundamental technique in machine learning and is part of most modern methods.



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Empirical Probability Distributions

Consider N trials $\{(X_1, Y_1), \dots, (X_N, Y_N)\}$ in which we sample (X, Y) .

Let n_{ij} be the number (count) of trials where $(X = x_i, Y = y_j)$, we can construct a discrete **empirical probability distribution**:

$$\hat{p}(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Frequentist View: In the limit as $N \rightarrow \infty$, the empirical distribution converges (with probability 1, by the law of large numbers) to the true distribution:

$$p(X = x_i, Y = y_j) = \lim_{N \rightarrow \infty} \hat{p}(X = x_i, Y = y_j)$$



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The Empirical Distribution Is a Model

The empirical distribution is a **model** of the probability of events:

$$\hat{p}(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

- It satisfies **non-negativity** and **normalization**, we can marginalize it, condition on it, and apply Bayes' theorem to it.

Is it a good model of the probability of events?

- What are some issues with this approach when N is **small**?
- What happens when there are **many variables**?
- What happens when the variables take on **many values** or are **continuous**?



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Estimating the Parameters of a Distribution

Given a collection of data $\mathcal{D} = \{x_1, \dots, x_N\}$, there are several ways to estimate the “best” parameters for a given distribution.

- **Maximum Likelihood Estimation (MLE):** choose the parameters that maximize the **likelihood** of the data. (Frequentist)

$$w_{\text{MLE}}^* = \arg \max_w p(\mathcal{D} \mid w)$$

- **Maximum A Posteriori (MAP) Estimation:** choose the parameters that maximize the posterior given the data and the **prior**. (Bayesian)

$$w_{\text{MAP}}^* = \arg \max_w p(\mathcal{D} \mid w) p(w)$$

Both methods rely on computing the **likelihood of the data**.

- We cover both today: MLE first, then MAP.



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The Likelihood Function

The likelihood function describes the **probability of the observed data** $\mathcal{D} = \{x_1, \dots, x_N\}$ under a particular distribution.

If we assume each observation $X_i \sim p(x|w)$ is an **Independent and Identically Distributed (IID)** sample from some parametric distribution $p(x|w)$ parameterized by w then the likelihood is:

$$p(\mathcal{D}|w) = \prod_{n=1}^N p(x_n | w)$$

Independent

Identically



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The Log Likelihood Function

In practice, we will typically work with the **log likelihood function**:

$$p(\mathcal{D}|w) = \prod_{n=1}^N p(x_n | w) \implies \ln p(\mathcal{D}|w) = \sum_{n=1}^N \ln p(x_n | w)$$

We take the **log** of the likelihood function for several reasons:

- It's **easier to manipulate summations** than products (e.g., calculus).
- The log probabilities and summation are **more numerically stable**.

Does this change our **optimization problem**?

- No – log is **monotonically increasing function** of its arguments:

$$\arg \max_w p(\mathcal{D}|w) = \arg \max_w \ln p(\mathcal{D}|w)$$



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Maximum Likelihood Estimation

Maximum likelihood estimation has two steps:

1. **Modeling:** Define the **log likelihood function** of the **data**: $\ln p(\mathcal{D}|w)$
2. **Optimization:** Compute the parameters that **maximize** the log-likelihood:

$$w_{\text{ML}} = \arg \max_w \ln p(\mathcal{D}|w)$$

Some machine learning people prefer to **minimize** a loss (same thing):

$$w_{\text{ML}} = \arg \min_w -\ln p(\mathcal{D}|w)$$

- Maximum likelihood estimation (MLE) $\rightarrow$ **minimize the log loss**.

Maximum likelihood estimation is a core technique in ML and is the basic objective for everything from the **logistic regression** to **ChatGPT**



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The MLE for ChatGPT

We model the likelihood of all the documents on the internet:

$$\begin{aligned} & \ln p(\text{Internet} \mid w) \\ &= \sum_{\text{Doc} \in \text{Internet}} \sum_{\text{Word} \in \text{Doc}} \ln p(\text{Word} \mid \text{Words before it in Doc}, w) \end{aligned}$$

Pre-Training: billions of dollars spent annually computing an MLE:

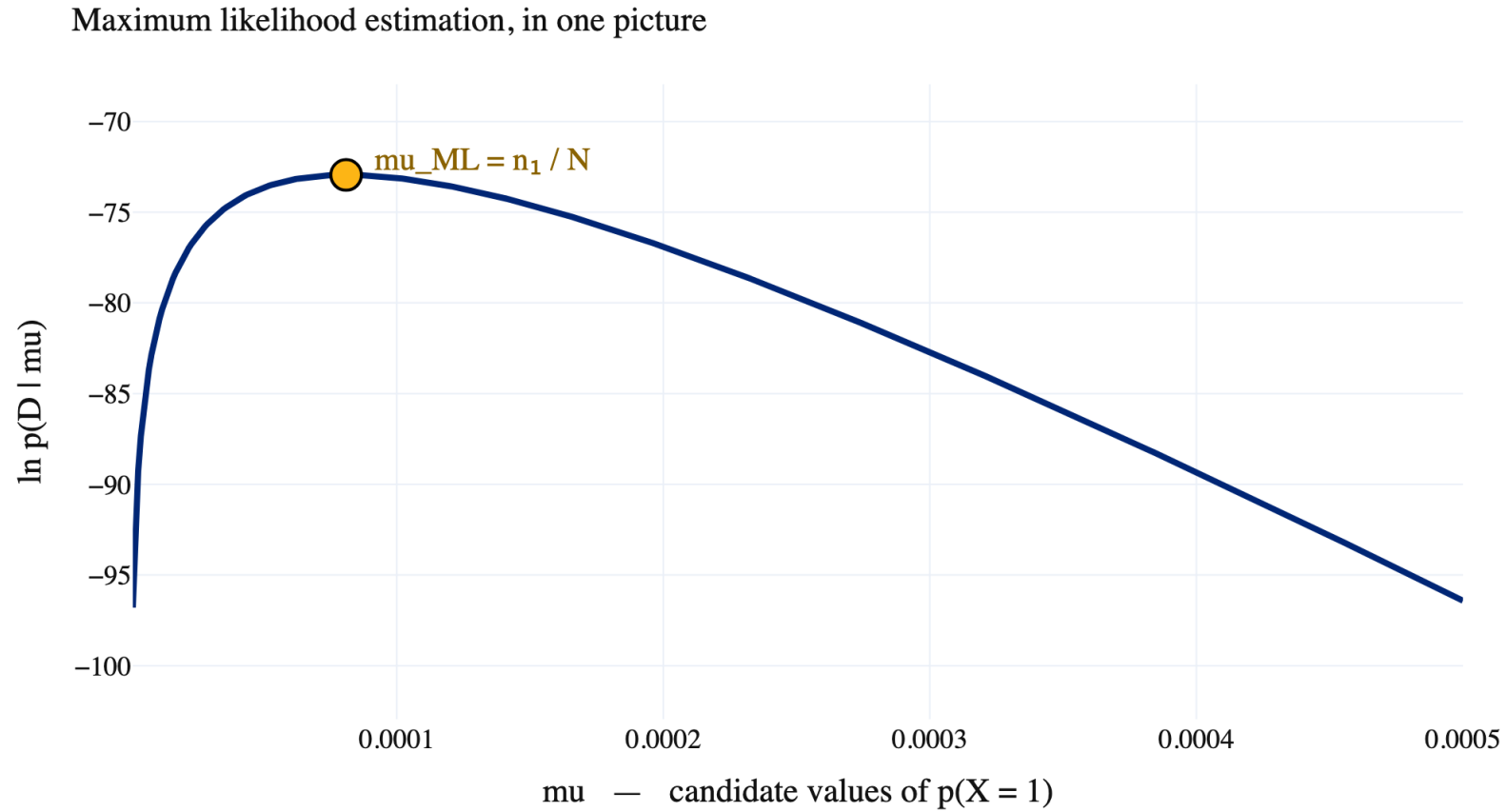
$$\arg \max_w \ln p(\text{Internet} \mid w)$$



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Demo

Maximum likelihood,
in one picture





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The MLE for IID Bernoulli Samples (Part 1)

Example: Bernoulli Distribution

We observe N samples $\mathcal{D} = \{x_1, \dots, x_N\}$ from a Bernoulli Distribution.

1. Modeling: Let's assume IID Bernoulli trials:

$$\ln p(\mathcal{D}|\mu) = \sum_{n=1}^N \ln(\mu^{x_n}(1-\mu)^{(1-x_n)}) = \sum_{n=1}^N x_n \ln(\mu) + (1-x_n) \ln(1-\mu)$$

Since each x_n is either **zero** or **one** we can rewrite this expression as:

$$\ln p(\mathcal{D}|\mu) = n_1 \ln(\mu) + n_0 \ln(1-\mu)$$

Where n_0 and n_1 are the number of observed zeros and ones, respectively.

The counts n_0 and n_1 are called **sufficient statistics** because they are all we need from the data to estimate the parameter.

The MLE for IID Bernoulli Samples (Part 2)



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2. Optimization: Maximize the log likelihood function:

$$\arg \max_{\mu} n_1 \ln(\mu) + n_0 \ln(1 - \mu)$$

Here we use the **stationary-points approach** to optimization:

1. Take the derivative $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu)$

$$\frac{\partial}{\partial \mu} (n_1 \ln(\mu) + n_0 \ln(1 - \mu)) = \frac{n_1}{\mu} - \frac{n_0}{1 - \mu}$$

2. Solve for the stationary point $\frac{\partial}{\partial \mu} \ln p(\mathcal{D}|\mu) = 0$

$$\frac{n_1}{\mu} - \frac{n_0}{1 - \mu} = 0 \implies n_1(1 - \mu) = n_0\mu \implies \mu_{ML} = \frac{n_1}{n_0 + n_1}$$

3. Verify that stationary points are maximum: $\frac{\partial^2}{\partial \mu^2} \ln p(\mathcal{D}|\mu) < 0$

Left as an
exercise
(for your AI).

Stopped Here



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The Bernoulli MLE Is Just Counting

The maximum likelihood estimate (MLE) is:

$$\mu_{ML} = \frac{n_1}{n_0 + n_1} = \frac{n_1}{N}$$

This is **the fraction of ones in the data**, which is the **empirical distribution** defined earlier in lecture.

➤ μ_{ML} is the parameter value that makes the **observed data most probable** under a Bernoulli model.

The same argument applied to the **tabular model** gives:

$$\hat{p}(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Therefore, not only are **counts** the intuitive estimate but they are **also the maximum likelihood estimate**.



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Your Turn: Let's flip a coin

I flip a coin $N = 10$ times and observe 7 heads. What is $\mu_{ML} = p(\text{heads})$

$$\mu_{ML} = \frac{n_1}{N} = \frac{7}{10}$$

I flip a different coin 10 times and observe 0 heads. What is $\mu_{ML} = p(\text{heads})$?

$$\mu_{ML} = \frac{n_1}{N} = \frac{0}{10}$$

Your model assigns **zero probability** to heads $p(\text{heads}) = 0$. Are you happy with that estimate?

- It is the correct maximum likelihood estimate.
- It is also almost certainly wrong! (Assuming my coin has two sides...)

Rare events produce confident but uninformative estimates. The remedy is to introduce a prior, which is **MAP estimation**.



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The Issue with Maximum Likelihood and Rare Events

The Bernoulli MLE $\mu_{ML} = \frac{n_1}{N}$ is the observed frequency.

➤ When the event is rare, the **observed frequency is often zero**.

With $N = 10,000$ segments and a true rate of 0.0001:

$$p(n_1 = 0) = (1 - 0.0001)^{10,000} \approx 0.37$$

we observe $n_1 = 0$ about a *third of the time*.

➤ A third of the time we would say wake words are never said.

We know something the data does not contain; wake words can occur.

➤ We can incorporate this using a **Maximum a Posteriori (MAP) estimate**.



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The Parameter as a Random Variable

Maximum likelihood estimation treats μ as an unknown constant.

The Bayesian view treats μ as a random variable with a distribution. Applying Bayes' theorem to the parameter:

$$p(\mu | \mathcal{D}) = \frac{p(\mathcal{D} | \mu) p(\mu)}{p(\mathcal{D})}$$

- **Prior** $p(\mu)$: what we believe about μ before seeing data.
- **Likelihood** $p(\mathcal{D} | \mu)$: is the quantity we maximized for the MLE.
- **Posterior** $p(\mu | \mathcal{D})$: the updated belief after seeing the data.

Maximum a Posteriori (MAP) estimate maximizes the posterior:

$$\mu_{\text{MAP}} = \arg \max_{\mu} \ln p(\mathcal{D} | \mu) + \ln p(\mu)$$

- The denominator $p(\mathcal{D})$ does not depend on μ , so it does not affect the maximization.



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Modeling the Prior

The **choice** and **parameterization** of model for the prior $p(\mu)$ is a **design decision**.

- Capture our prior beliefs about the μ .
- Model is often chosen for convenience (conjugate to the likelihood).
- The prior parameters are set using *prior knowledge* or *another prior*.

What would be a good prior $p(\mu)$ for the Bernoulli coin flip example?

- The parameter $0 \leq \mu \leq 1$ is the probability of landing heads
- We expect the coin (and the flipper) to be approximately fair.



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The Beta Distribution

The **Beta distribution** for the **continuous random variable** $0 \leq X \leq 1$:

$$p(X; a, b) = \frac{1}{B(a, b)} X^{a-1} (1 - X)^{b-1}$$

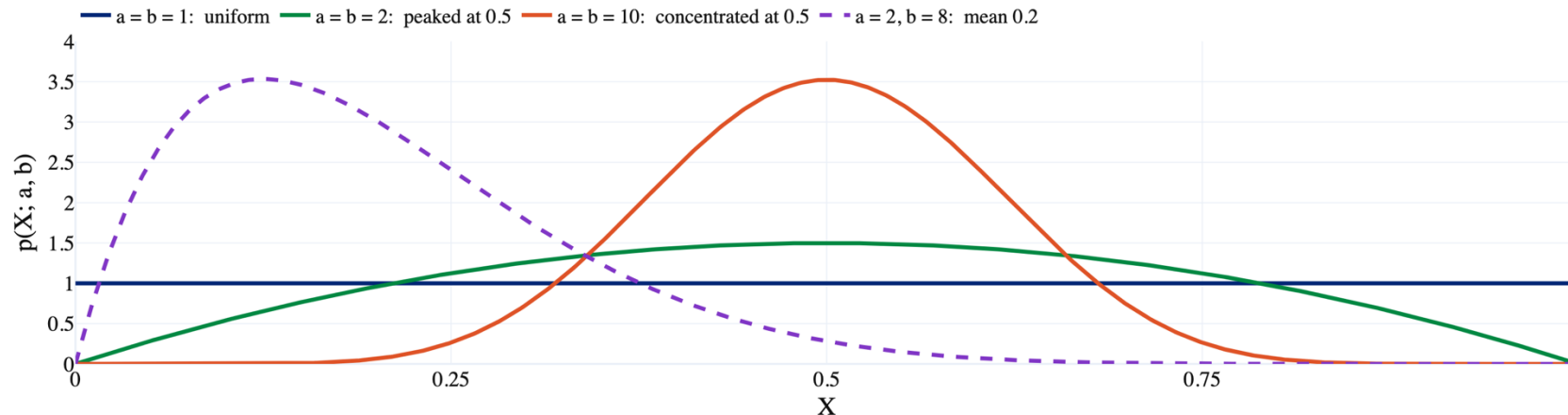
with **parameters by α and β** with the following shape:

➤ $a = b = 1$: every value of X is equally plausible.

➤ $a = b = 2$: values of X close to 0.5 are more likely

➤ Large a and b concentrate the distribution near $\frac{a}{a+b}$

The beta function $B(a, b)$ is the normalizer (computed numerically) and outside the scope of this class.





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Deriving the Posterior for Bernoulli + Beta

Recall we are trying to compute:

$$p(\mu | \mathcal{D}) = \frac{p(\mathcal{D} | \mu) p(\mu)}{p(\mathcal{D})}$$

Substituting the Bernoulli and the Beta distributions (without normalization):

$$p(\mu | \mathcal{D}) \propto p(\mathcal{D} | \mu) p(\mu) = \mu^{n_1} (1 - \mu)^{n_0} \mu^{(a-1)} (1 - \mu)^{(b-1)}$$

Collecting terms, we can see that the **posterior is also a Beta distribution**:

$$p(\mu | \mathcal{D}) \propto \mu^{(n_1+a-1)} (1 - \mu)^{(n_0+b-1)}$$

with parameters $a' = (n_1 + a)$ and $b' = (n_0 + b)$.

When a prior paired with a likelihood has the same form as the posterior it is called a **conjugate prior**.

➤ The Beta distribution is the conjugate prior for the Bernoulli likelihood.



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Computing the MAP

Maximum a Posteriori (MAP) estimate maximizes the posterior:

$$\mu_{\text{MAP}} = \arg \max_{\mu} \ln p(\mathcal{D} | \mu) + \ln p(\mu)$$

When deriving the posterior we already showed (collecting terms):

$$\ln p(\mathcal{D} | \mu) + \ln p(\mu) = (n_1 + a - 1) \ln \mu + (n_0 + b - 1) \ln(1 - \mu)$$

When computing the MLE we derived the **stationary point** of:

$$\frac{n_1}{n_1 + n_0} = \arg \max_{\mu} n_1 \ln \mu + n_0 \ln(1 - \mu)$$

Substituting $n'_1 = (n_1 + a - 1)$ and $n'_0 = (n_0 + b - 1)$.

$$\mu_{\text{MAP}} = \frac{n_1 + a - 1}{n_1 + n_0 + a + b - 2} = \frac{n_1 + a - 1}{N + a + b - 2}$$



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The Prior as Pseudo-Counts

The prior acts like a form of pseudo-counts (assumed binary observations) by contributing $a - 1$ imaginary successes $b - 1$ imaginary failures.

$$\mu_{\text{MAP}} = \frac{n_1 + a - 1}{n_1 + n_0 + a + b - 2} = \frac{n_1 + a - 1}{N + a + b - 2}$$

With $a = b = 2$ this becomes $\mu_{\text{MAP}} = \frac{n_1 + 1}{N + 2}$ one imaginary observation of each outcome.

➤ Regardless of the observations $0 < \mu_{\text{MAP}} = \frac{n_1 + 1}{N + 2} < 1$ the probability is never 0 or 1.

This is also called **Laplace Smoothing**

➤ Add **pseudo counts to empirical estimates** to act as a prior when there is **limited data** or **events are rare**.



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How μ_{MAP} and μ_{ML} scale with N

What happens as N grows and how do μ_{MAP} and μ_{ML} compare?

$$\mu_{\text{ML}} = \frac{n_1}{n_0 + n_1} = \frac{n_1}{N}, \quad \mu_{\text{MAP}} = \frac{n_1 + a - 1}{n_1 + n_0 + a + b - 2} = \frac{n_1 + a - 1}{N + a + b - 2}$$

As N grows, the constants $a - 1$ and $a + b - 2$ become negligible and the **two estimates converge**.

- **Small N :** the prior dominates, and the estimate is pulled toward the prior mean.
- **Large N :** the data dominates, and MAP and MLE agree.



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What We Did Today

Reviewed basic probability, Bayes Theorem, Expectations, and Variance

Maximum likelihood determine model parameters by maximizing the likelihood of the data under the model.

Maximum a posteriori determine model parameters that maximize the posterior distribution.

Next Lecture: we will combine ideas learned today to develop Gaussian Mixture Models.

Lecture 4

Probability and Density Estimation

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters:

- **Probability:** Chapter 2.[1-2]
- **Density Estimation and MLE:** Chapter 2.3